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Heat Kernel Meet Geodesics: A Hybrid Approach to 3d Voronoi Meshing for Reservoirs with High Permeability Zone

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Abstract

Mesh generation is one of the most critical parts for numerical pressure transient analysis to ensure the simulation efficiency. Perpendicular Bisection (PEBI) grid, a kind of Voronoi grid, is applied because of its flexibility, accuracy representation of anisotropy and so on. However, most of algorithm is designed for 2D and 2.5D (structured grid at vertical direction) situation. They inadequately resolve 3D pressure dynamics near wellbores, interlayer interfaces, and complex boundaries. Some commercial software like Kappa™ try to combine 3D Voronoi grid which around wellbores with background 2D grid system. But this method compromises conciseness and significantly increases programming complexity. To solve these problems, this paper proposes an improved 3D Voronoi grid generation method which based on Riemannian geometry and heat equation. The primary improvement of this method is to optimize the control points to build 3D Voronoi grid. In this method, each completion segment emits geodesic lines spherically, which reflect or refract at boundaries according to prescribed conditions. After generating geodesic lines for all point sources, pressure drop distributions along these paths are calculated using thermal kernel functions based on assigned flow rates. Discrete points with varying densities are then distributed along the geodesic lines according to the pressure gradient magnitudes. These points serve as control nodes for Voronoi grid generation. The resulting 3D PEBI grid exhibits superior adaptive resolution across spatial scales. Near wellbores where pressure gradients are steepest, densely packed Voronoi cells capture intricate flow patterns through their tetrahedral morphology, conforming precisely to the curvature of complex well architectures. This innovative 3D Voronoi framework integrates geodesic-driven thermal kernel discretization with adaptive node density modulation, which can be used for vertically heterogeneous reservoirs in Middle East.

Keywords: pressure transient analysis, Voronoi grid, Riemannian geometry, heat kernel

Introduction

Pressure transient analysis (PTA) is crucial in reservoir engineering but faces significant challenges in heterogeneous reservoirs or with complex well types (e.g., horizontal or multilateral wells). Consequently, numerical PTA has become a powerful tool, particularly when measured log-log plots deviate from analytical solutions and multi-phase flow is involved. As the foundational step of numerical PTA, grid generation is paramount for ensuring the convergence and accuracy of simulation results. Numerically, PTA can be viewed as a specialized form of reservoir simulation employing extremely small-time steps. While various grid types, including Cartesian and corner-point grids, are theoretically applicable, the application of PEBI (Perpendicular Bisection) grids – a type of Voronoi grid – represents a significant advancement. PEBI grids offer enhanced accuracy and efficiency for modeling complex flow scenarios in hydrocarbon reservoirs. While fundamentally Voronoi, PEBI grids possess unique features compatible with reservoir transient behavior. For instance, five primary PEBI grid types exist: variable-density, circular, rectangular, elliptical, and hexagonal^[1]. Depending on well type and boundary conditions, different grid types are often combined to model the tested reservoir sector. Commercial software like Saphir™ typically employs a hybrid strategy for PEBI grid generation. This strategy divides grid construction into two main components: background grids and refined grids. Background grids, usually 2D or 2.5D rectangular or hexagonal, describe bulk properties based on reservoir geometry. Grids near wellbores are then refined using 2D or 3D grids to capture early-time pressure behavior after shut-in^[2]. This approach balances accuracy with generation complexity, mitigating non-convergence issues caused by excessively rapid changes in PEBI grid size. However, this method presents coding challenges and has limited capability for accurately depicting boundaries in three-dimensional models. Addressing these limitations necessitates an improved 3D Voronoi grid generation method.

Previous research on PEBI grids, particularly within reservoir numerical simulation, provides significant reference value. As discussed, a critical step in building a PEBI grid system is defining the distribution of nodes (grid center points) around the wellbore, including their positions and density. For example, around a fully penetrating vertical well, nodes are distributed radially, with density increasing logarithmically towards the wellbore. This density is often calculated using power-law equations derived from stable planar radial flow theory^[3]. This approach inherently incorporates flow path considerations to capture stream field characteristics. Some research has even introduced streamline simulation concepts to generate PEBI grids, specifically optimizing flow simulation for patterns like the five-spot well configuration^[4]. This work is highly instructive because the underlying concept of streamlines based on time of flight (TOF) – the time required for a pressure wave front to propagate a unit length – has profound mathematical foundations. Essentially, spatial derivatives of the TOF reflect local pressure transmission capabilities^[5]. From the perspective of intrinsic geometry, this primarily manifests as variations in the metric tensor (a core concept in Riemannian geometry) across the reservoir domain.

While Riemannian geometry is rarely explicitly mentioned in reservoir engineering, it finds widespread application in thermodynamics^[6]. Building on geodesics (the generalized concept of straight lines in Riemannian manifolds), transformation thermodynamics was developed. In this field, scientists manipulate the metric tensor at different positions to control the direction of heat flow^[7-13]. Similarly, the concept of heatlines, analogous to streamlines, was proposed for heat flow analysis^[14-16]. Given the mathematical similarity between single-phase seepage equations and heat conduction equations, methods like heatlines and heat kernel functions can also be leveraged for pre-analysis prior to flow simulation^[17-19]. Results from such pre-analysis can guide the generation of higher-quality meshes capable of capturing critical temporal changes in the flow field.

This paper proposes an improved method for generating three-dimensional Voronoi meshes (PEBI grids). First, starting from coordinate transformation, we elucidate the fundamental application of Riemannian geometry to single-phase seepage problems. Next, building on concepts of geodesics, streamlines, and

heatline theory, we propose a method for the *a priori* analysis of single-phase flow. To validate this method's effectiveness, we analyze common PTA flow regimes such as planar radial flow, spherical flow, and linear flow. Subsequently, we employ the heat kernel function to determine control point density along geodesics, obtaining the control points for the PEBI mesh across the computational domain. Finally, Voronoi meshes are generated based on these control points. Results demonstrate that the proposed method exhibits strong adaptability to various well types and effectively captures flow field distortions caused by property variations across interfaces. This advancement has significant implications for numerical well testing. Furthermore, the relative simplicity of the underlying theory and methodology reduces implementation complexity, facilitating the development of new numerical well testing software and modules.

Intrinsic View of Flow Behavior and Geodesics Theory

Mathematical Deduction of Single-Phase Flow Equation Based on Riemannian Geometry

Considering the anisotropy between horizontal and vertical permeability, the three-dimensional single-phase flow equation can be written as:

$$k_h \frac{\partial^2 p}{\partial x^2} + k_h \frac{\partial^2 p}{\partial y^2} + k_v \frac{\partial^2 p}{\partial z^2} = \phi \mu C_t \frac{\partial p}{\partial t} \quad (1)$$

Define the ratio variable of vertical-to-horizontal permeability as follows:

$$\beta = \frac{k_v}{k_h} \quad (2)$$

Thus, the flow equation can be reformulated as:

$$k_h \frac{\partial^2 p}{\partial x^2} + k_h \frac{\partial^2 p}{\partial y^2} + \beta k_h \frac{\partial^2 p}{\partial z^2} = (\phi \mu C_t) \frac{\partial p}{\partial t} \quad (3)$$

Let $\eta = \frac{k_h}{\phi \mu C_t}$, $Z = \sqrt{\beta} z$, then the equation transforms into an isotropic model in three dimensions.

$$\frac{\partial^2 p}{\partial x^2} + \frac{\partial^2 p}{\partial y^2} + \frac{\partial^2 p}{\partial Z^2} = \frac{1}{\eta} \frac{\partial p}{\partial t} \quad (4)$$

Without loss of generality, let $X = x$, $Y = y$, yielding:

$$\frac{\partial^2 p}{\partial X^2} + \frac{\partial^2 p}{\partial Y^2} + \frac{\partial^2 p}{\partial Z^2} = \frac{1}{\eta} \frac{\partial p}{\partial t} \quad (5)$$

This represents the commonly used coordinate transformation method in flow mechanics. Through this approach, the governing equation achieves its most concise form for subsequent solving. However, this single governing equation applies only to homogeneous reservoirs (when the ratio of vertical and horizontal permeability is a constant value, vertical heterogeneity can be neglected as it can be resolved through processing methods). When permeability varies across different regions or layers, the governing equations for distinct zones also change according to variations in permeability, porosity, and rock compressibility. For clarity, we will discuss in detail the scenario where the entire reservoir comprises only two regions with differing permeabilities, though the same principles apply to other cases. The governing equations for different regions can be expressed as:

$$k_{h1} \frac{\partial^2 P_1}{\partial x^2} + k_{h1} \frac{\partial^2 P_1}{\partial y^2} + \beta k_{h1} \frac{\partial^2 P_1}{\partial z^2} = (\phi \mu C_t)_1 \frac{\partial P_1}{\partial t} \quad (6)$$

$$k_{h2} \frac{\partial^2 P_2}{\partial x^2} + k_{h2} \frac{\partial^2 P_2}{\partial y^2} + \beta k_{h2} \frac{\partial^2 P_2}{\partial z^2} = (\phi \mu C_t)_2 \frac{\partial P_2}{\partial t} \quad (7)$$

Set $\beta_1 = \frac{k_{v1}}{k_{h1}}$, $\beta_2 = \frac{k_{v2}}{k_{h2}}$, assuming the ratio of vertical and horizontal permeability in the oil layer is similar, then:

$$\frac{\partial^2 p}{\partial x^2} + \frac{\partial^2 p}{\partial y^2} + \beta \frac{\partial^2 p}{\partial z^2} = \frac{1}{\eta_1} \frac{\partial p}{\partial t} \quad (8)$$

$$\frac{\partial^2 p}{\partial x^2} + \frac{\partial^2 p}{\partial y^2} + \beta \frac{\partial^2 p}{\partial z^2} = \frac{1}{\eta_2} \frac{\partial p}{\partial t} \quad (9)$$

To unify the right-hand sides of both equations, transform the equation for Region 2:

$$\frac{\partial^2 p}{\partial x^2} + \frac{\partial^2 p}{\partial y^2} + \beta \frac{\partial^2 p}{\partial z^2} = \frac{1}{\eta_1} \frac{\partial p}{\partial t} \quad (10)$$

$$\frac{\eta_2}{\eta_1} \left(\frac{\partial^2 p}{\partial x^2} + \frac{\partial^2 p}{\partial y^2} + \beta \frac{\partial^2 p}{\partial z^2} \right) = \frac{1}{\eta_1} \frac{\partial p}{\partial t} \quad (11)$$

For Region 2, introduce the following coordinate transformation:

$$\begin{bmatrix} \sqrt{\frac{\eta_1}{\eta_i}} & & \\ & \sqrt{\frac{\eta_1}{\eta_i}} & \\ & & \sqrt{\frac{\eta_1}{\eta_i}} \end{bmatrix} \begin{bmatrix} 1 & & \\ & 1 & \\ & & \sqrt{\beta} \end{bmatrix} \begin{bmatrix} x \\ y \\ z \end{bmatrix} = \begin{bmatrix} X^* \\ Y^* \\ Z^* \end{bmatrix} (i = 1, 2) \quad (12)$$

For Region 1, the coordinate transformation remains consistent with the homogeneous case. At this point, the equations for both regions can be unified as:

$$\frac{\partial^2 p}{\partial X^{*2}} + \frac{\partial^2 p}{\partial Y^{*2}} + \frac{\partial^2 p}{\partial Z^{*2}} = \frac{1}{\eta_1} \frac{\partial p}{\partial t} \quad (13)$$

From a physical perspective, hydraulic diffusivity represents the area covered by a pressure wave per unit time. Thus, dimensionally, the ratio of hydraulic diffusivities represents the proportional area covered by pressure waves per unit time. Similarly, the square root of this ratio corresponds to the distance traveled by the pressure wave per unit time. Evidently, differences in pressure wave propagation speed can be equivalently described as disparities in displacement per unit time. That is, from the perspective of the pressure wave itself, the distance it propagates within different reservoir properties varies per unit time. From the viewpoint of intrinsic geometry—distinct from the external observer's perspective (typically Eulerian perspective)—different regions exhibit distinct coordinate scaling rules. The Jacobian matrices for coordinate transformations in different regions can be expressed as:

$$J = \begin{bmatrix} \sqrt{\frac{\eta_1}{\eta_i}} & & \\ & \sqrt{\frac{\eta_1}{\eta_i}} & \\ & & \sqrt{\beta} \sqrt{\frac{\eta_1}{\eta_i}} \end{bmatrix} \quad (14)$$

Therefore, according to the definition of the metric tensor in Riemannian geometry, the metric tensor for pressure propagation in different regions can be written as:

$$\bar{G} = [g^{ij}] = JJ^T \quad (15)$$

It thus becomes evident that the pressure wave propagation process in the single-phase flow equation can be described using Riemannian geometry. After transformation, the mathematical model seems like an entire uniform reservoir. But it should be noticed that physical property variations in oil reservoirs are

often discontinuous; thus, the entire reservoir cannot be equivalently modeled as a complete, compact, and continuous Riemannian manifold.

The theory of geodesics underlying the fundamental solution of equations

In prior research on grid systems for numerical well testing, a variety of geometric modules have been employed, including—but not limited to—points, lines, circles, ellipses, and planes. As illustrated in Table 1, distinct geometric elements correspond to different reservoir components. Accordingly, the selected element influences the grid configuration. For example, when representing the near-well region of a vertical well, circular modules typically utilize radial grids. These grids are centered on the wellbore and extend outward in concentric rings, enabling accurate characterization of radial flow behavior around the wellbore^[20].

Table 1—Geometrical Elements and Corresponding Reservoir Units.

Element	Reservoir Units
Points	• Vertices of the boundary polygon: starting points, inflection points • Key points of faults • Intersections between faults • Intersections between faults and the boundary
Lines	• Neighborhood of boundary and faults excluding point modules
Circles	• Neighborhood of vertical well
Ellipses	• Neighborhoods of fractured Vertical wells and horizontal wells
Planes	• Top and bottom surface of the reservoir and other boundaries

Traditional polygonal grids, such as orthogonal grids, require dense refinement near the wellbore to approximate circular boundaries. This often introduces stepped geometric artifacts, which can distort flow direction and reduce simulation fidelity. Consequently, numerical well testing frameworks often adopt hybrid grid systems. The fundamental premise of such hybridization lies in leveraging prior knowledge of the flow field geometry to optimize grid partitioning, thereby enhancing the resolution of pressure and saturation gradients.

Despite their hybrid nature, these grids fundamentally conform to the structure of Voronoi grids—commonly referred to as PEBI (Perpendicular Bisector) grids. As a generalized form of point-distributed grids, Voronoi grid generation algorithms can produce diverse grid configurations by adjusting the spatial distribution and density of control points, offering substantial flexibility in capturing complex reservoir geometries (Fig. 1).

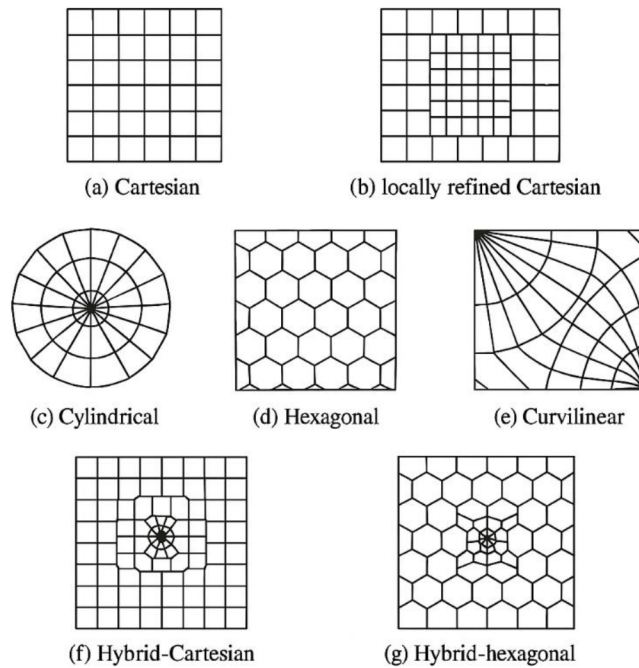


Figure 1—Different cases of Voronoi grid with different point distribution^[20]

For a point source in an infinite homogeneous space, the fundamental solution of the flow equation is:

$$p_i - p(\vec{r}, t) = \frac{Qu}{(4\pi\eta t)^{\frac{3}{2}}} e^{-\frac{|\vec{r}|^2}{4\eta t}} \quad (17)$$

Obviously, this solution is similar with the fundamental solution of heat equation due to same governing equation. For fundamental solution which is also called "heat kernel function" or "Weierstrass function", there is an effective method named "Varadhan's formula" to calculate the distance of the propagation of heat contour. According to the calculation of the limitation which is shown in [Appendix A](#), in an unbounded homogeneous medium, the pressure wavefront takes the form of a spherical surface. At every point on this surface, propagation follows a straight line radiating outward along the sphere's radius—i.e., a geodesic in the case of a constant metric tensor. Therefore, for a point source in three-dimensional space, geodesic paths can be uniformly sampled throughout the domain, and control points placed along each path. By leveraging these control points, PEBI grids that conform to the a priori flow behavior can be generated. For grid system partitioning around a point source, geodesic lines may be radially extended from the source in multiple directions. Control points are subsequently determined along these geodesics according to prescribed density distributions. The well-established Voronoi algorithm can then be applied to these control points to generate perpendicular bisector (PEBI) grids that conform to flow priori knowledge. This methodology has been validated through simulations of fully perforated vertical wells in infinite reservoirs (mathematically equivalent to a point source in 2D space) ([Fig. 2](#)). Completion segments of various well types can be mathematically represented as line integrals of point sources along well trajectories. Consequently, it appears theoretically feasible to generate geodesics and control points through point source superposition. However, significant discrepancies between theoretical models and practical implementations render direct control point construction from discrete point sources inadvisable, primarily due to:

- Individual point sources require excessive control points to generate PEBI grid clusters approximating spherical geometries. This stems from the three-dimensional constraints inherent in spherical coordinate systems during geodesic generation around each source ([Fig 3](#)).

- Spatial discontinuities between discrete point sources cause intersection and interference of geodesics from adjacent sources, leading to anomalous density distributions and localized grid distortions between sources.
- Spherical Uniform Distribution Problem is a difficult mathematical problem, which sometimes require complicated calculation or physical-informed simulation. The simplest solution to this problem has their shortcomings:

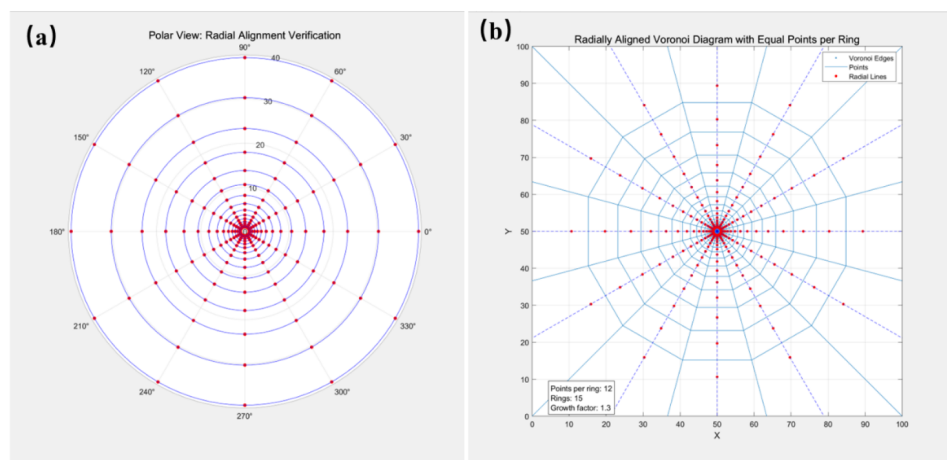


Figure 2—Test of Voronoi grid generation of a vertical well (a) The designed point distribution (b) Generated Voronoi grid

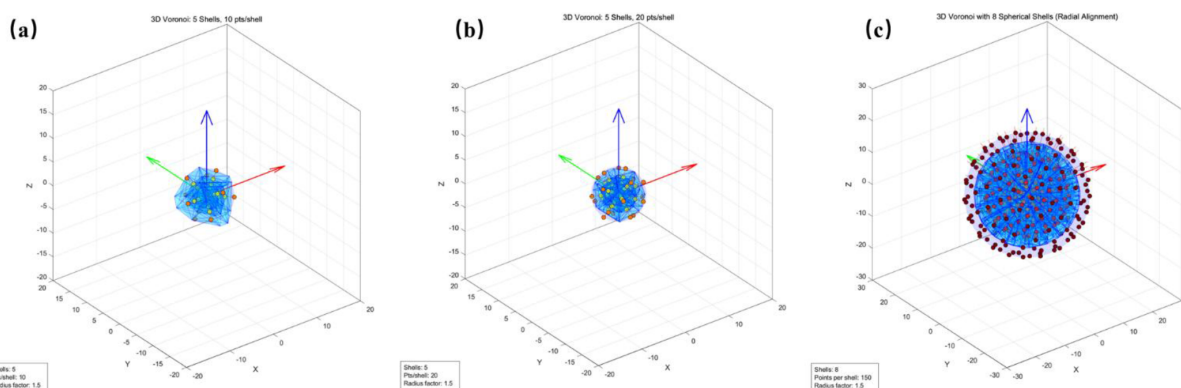


Figure 3—3d Voronoi grid generation results by different number of control points: (a) 50 points; (b) 100 points; (c) 150 points

Equiangular Spherical Partitioning: Requires angular step increments to be exact divisors of 360° , fundamentally restricting precise control over geodesic line quantities.

Fibonacci Spherical Sampling: While suitable for generating predetermined quantities of geodesics, this method exhibits extreme sensitivity to scaling factor precision. Both excessive and insufficient parameter values cause distribution distortions.

Methodology of 3d Voronoi Grid Generation

It is evident that directly combining discrete classical point-source solutions to obtain the geodesic distribution along the entire completion interval is not a sound approach (analogous to numerical analysis, where one would not attempt to compute a function's integral by simply summing discrete function values). Therefore, it is necessary to attempt the averaging and equivalence of an infinite number of point sources distributed along the completion interval. This process yields a geometric surface within the vicinity of the completion segment and its corresponding geodesic distribution.

The essence of geodesic tracing involves selecting a closed surface within the neighborhood surrounding the completion interval. After uniformly sampling points on this surface, geodesics are traced by projecting rays originating from these sample points along the direction of the surface normal vectors. Consequently, an equivalent surface surrounding the line source (composed of infinite point sources) must be derived. This concept can be analyzed by analogy with the flow behavior patterns observed in horizontal wells. For a horizontal well situated within an infinite reservoir, its flow behavior can be approximately decomposed into two directions: planar linear flow and planar radial flow. Hence, it is commonly approximated by an ellipsoidal or a capsule-shaped surface^[21]. While the ellipsoidal surface exhibits continuously evolving geometric characteristics over time steps (progressing from an initial straight line towards a circle), the capsule-shaped surface is generally preferred for flow analysis. Similarly, for a completion segment within an infinite space, it can be equivalently characterized by a capsule-shaped surface. This surface consists of hemispherical caps at both ends to capture the characteristics of spherical flow in space, connected by a cylindrical midsection to capture the characteristics of radial or linear flow.

After characterizing the completion section using a capsule body, points are uniformly generated on the capsule surface. Subsequently, geodesic lines are drawn in the direction of the normal vector of the capsule surface at each sampling point, extending until the geodesic intersects a boundary (such as the reservoir geometry boundary, faults, fractures, etc.). Then, the distribution of control points along the geodesic line is calculated. Starting from the surface sampling point, the density of control points gradually decreases, and its density is controlled by a power-law distribution based on the approximate expansion of the heat kernel function. After generating the control points, a Delaunay tetrahedralization is performed, followed by the generation of a three-dimensional Voronoi mesh (Fig. 5). Once the mesh is generated, its quality is checked, problematic sections are corrected, and finally, it is output as a grid system suitable for numerical pressure transient analysis.

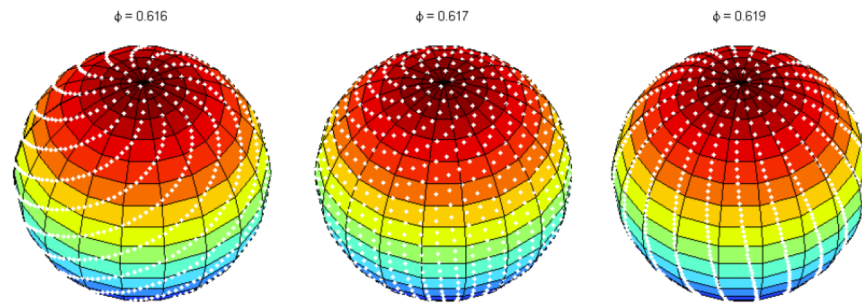


Figure 4—Different results of point distribution of Fibonacci spherical sampling

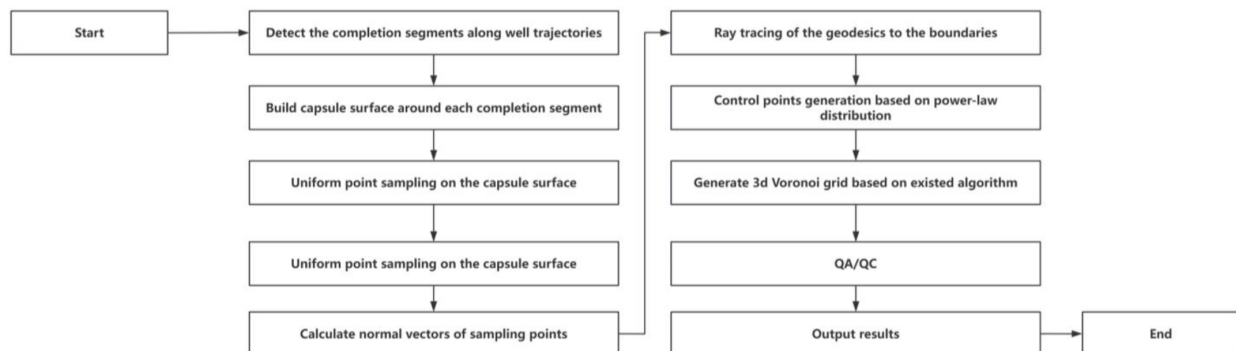


Figure 5—Flow chart of 3d Voronoi grid generation method based on geodesics

Results & Discussion

For carbonate reservoirs in the Middle East, vertical heterogeneity is primarily manifested in the distribution of high-permeability streaks. Generally speaking, high-permeability streaks are widely distributed in the planar view, resulting in extremely high vertical heterogeneity. At this time, the applicable conditions for analytical solutions of well test interpretation cannot be satisfied, leading to significant deviations between calculated data and measured data (Fig. 6). Therefore, numerical well test models need to be used for calculation. However, generally speaking, the grids in the model generally maintain a uniform orthogonal distribution vertically, which ignores a situation where, when there are differences in permeability among multiple small layers, abrupt changes in horizontal seepage velocity exist at the interfacial surfaces between layers (Fig. 7). Conventional 3d Voronoi grid generation methods usually use rigid ellipsoidal modules or capsule modules to cover the entire computational domain and generate grids, so it is difficult to reflect this difference vertically. But for the geodesic method, due to the differences in transmissibility coefficients between layers, geodesics will exhibit phenomena similar to light refraction at the interlayer interfaces, so the grid system it depicts is more consistent with prior knowledge of reservoir flow.

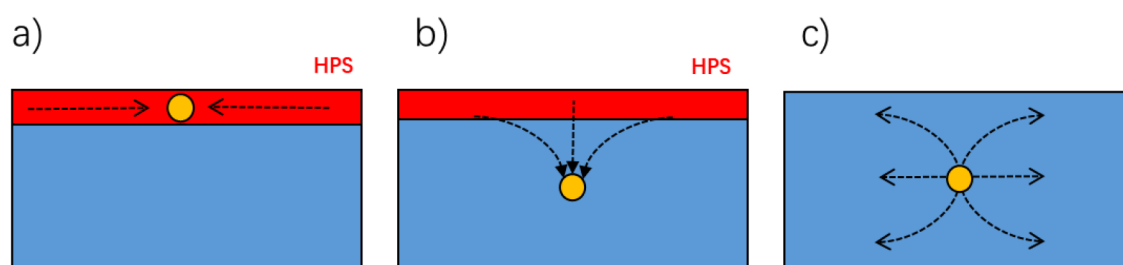


Figure 6—Scenario of different flow pattern of carbonate reservoir with high permeability streaks: (a) Streamline of horizontal section inside HPZ; (b) Streamline of horizontal section below HPZ; (c) Streamline of horizontal section inside a homogeneous reservoir.

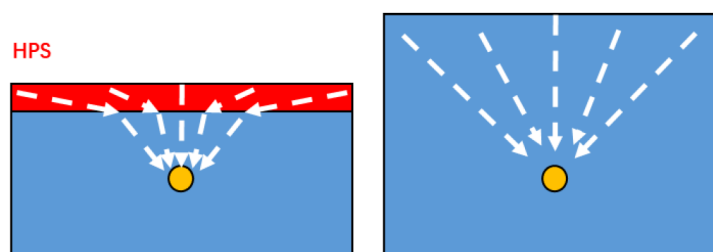


Figure 7—Illustration of flow speed abruptness by high permeability streaks.

Using the new grid generation algorithm to generate grids and perform simulations, the simulation results show that in the presence of a top high-permeability layer, fluids will mainly migrate along the high-permeability layer (Fig. 8). Even if the well trajectory is located outside the high-permeability layer, the fluid will also enter and exit the high-permeability layer through intense vertical flow, which is consistent with the resistivity logging results of the through-well.

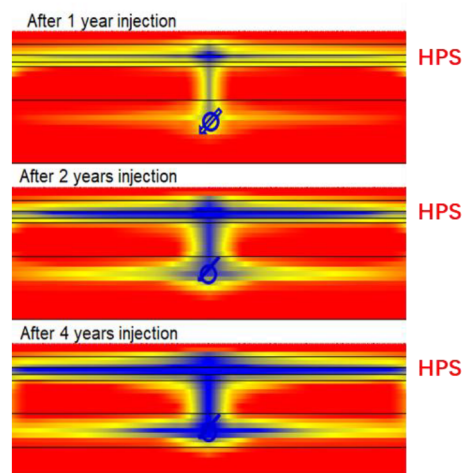


Figure 8—Flow direction of injected water with HPZ.

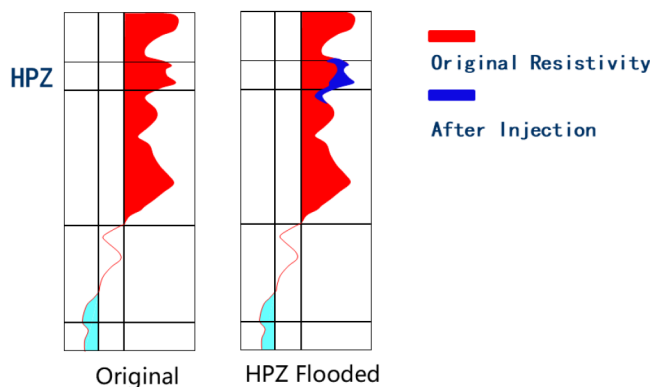


Figure 9—Comparison of resistivity logging results in reservoir A.

For horizontal well which has been applied mostly in Middle East. The slope of linear is nearly a quarter instead of a half unit slope due to high permeability zone (Fig 10). In log-log plot, it can be found that when reservoir is homogeneous, there is a half unit slope in log-log plot. With increasing permeability ratio of HPZ and other layers, a quarter unit slope occurs. Similarly, when horizontal is located at HPZ completely, streamlines concentrate on high permeability zone dominantly. Similarly, if well trajectory penetrates HPZ partially, the interval which is at HPZ will have the most production/intake rate due to drastic difference of reservoir permeability. Therefore, the log-log plot will be identical with analytical solutions.

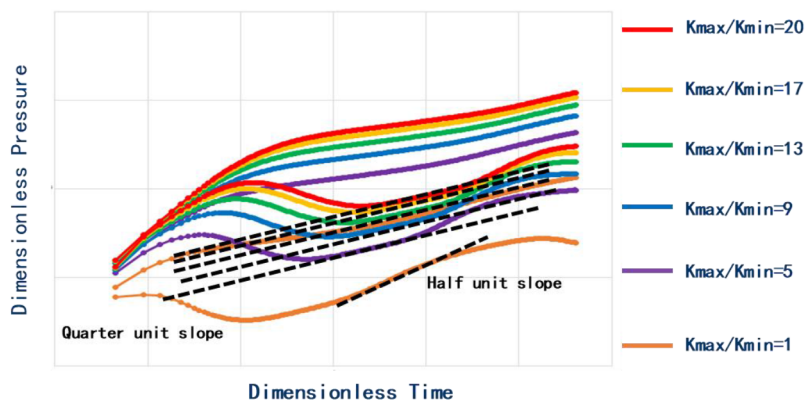


Figure 10—Sensitivity analysis of ratio of permeability between HPZ and other layers.

Take an injector I184 as an example, this well commissioned at 2013 whose horizontal trajectory below high permeability layer (Fig. 11). The log-log plot shows that there is a significant quarter unit slop during linear flow section. When using typical grid system, the interpreted effective horizontal length is only 299ft while total horizontal section is 2624ft. By new grid system, a better curve fitting and more reasonable results can be obtained (Table 2&Fig. 11).

Table 2—Comparison of interpreted results between two models of I184.

Parameters	Conventional Grid	Improved Grid
Skin Factor	-3.66	0.21
Effective Horizontal Length, ft	299.20	2624.67
Reservoir Permeability, mD	1.53	4.87
Kv/Kh	1.00	1.00
Thickness, ft	77.10	252.62
Height above Bottom Surface, ft	37.25	77.10

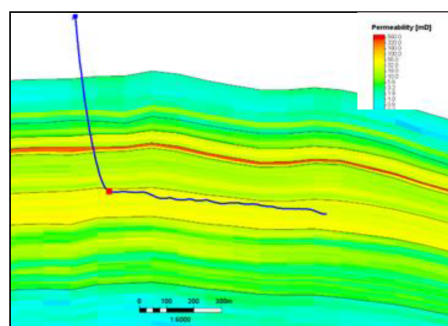


Figure 11—Trajectory and reservoir section of I184.

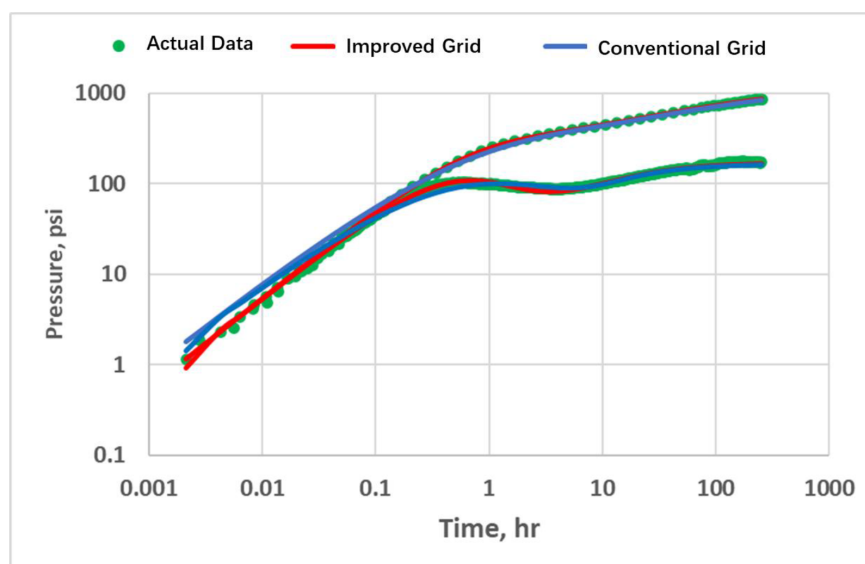


Figure 12—Comparison of curve fitting between two models of I184.

Conclusions and way forward

1. Previous PEBI grid generation methods are special cases of control points generation based on Riemannian geometry under homogeneous behavior. Mathematical proof has been proposed that the essence of 3d Voronoi grid generation is control points generation based on geodesics based on Riemannian geometry.
2. The improved 3d Voronoi grid generation method captures distortions in the flow field arising from heterogeneous properties at interfaces. High permeability layers play key factors which control pressure transient behavior of horizontal well. In this situation, the linear flow period will be a quarter slope instead of a half-unit slope.
3. The inherent simplicity of the theoretical framework and methodology reduces implementation complexity, thereby facilitating the development of new software and modules for numerical well testing.
4. In the future, more complex situation will be considered to build a more powerful algorithm which can fit different condition such as aquifer, partial leakage fault and so on.

Nomenclature

C_t	— Total compressibility, MPa ⁻¹
$\overline{G}$	— Metric Tensor
J	— Jacobi matrix of coordinate transformation
k_h	— Horizontal permeability, md
k_v	— Vertical permeability, md
p	— Reservoir pressure, MPa
p_i	— Initial reservoir pressure, MPa
$p(\vec{r}, t)$	— Reservoir pressure at specific time and position, MPa
Q	— Flow rate of the point source, m ³ /s
$\vec{r}$	— Position vector
t	— time, s
x	— x coordinate, m
y	— y coordinate, m
z	— z coordinate, m
X	— Transformed X coordinate, m
Y	— Transformed Y coordinate, m
Z	— Transformed Z coordinate, m
X^*	— Riemannian transformed X coordinate, m
Y^*	— Riemannian transformed Y coordinate, m
Z^*	— Riemannian transformed Z coordinate, m
β	— Ratio variable of vertical-to-horizontal permeability, dimensionless
ϕ	— Porosity, fraction
η	— Conductivity, m ² /s
μ	— Fluid viscosity, cp

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Appendix A

Proof of Varadhan's Formula for the Fundamental Solution of Single-Phase Flow

The fundamental solution of a point source in infinity homogeneous space with constant flow rate could be written as:

$$p_i - p(\vec{r}, t) = \frac{1}{(4\pi\eta t)^{\frac{3}{2}}} e^{-\frac{|\vec{r}|^2}{4\eta t}} \quad (\text{A-1})$$

Vector $\vec{r}$ is a vector representing a given point, defined as:

$$\vec{r} = [x - x_0, y - y_0, z - z_0]^T \quad (\text{A-2})$$

According to Varadhan's Formula, the geodesic distance between the certain point and the position of the point source ϕ could be calculated as:

$$\phi^2 = \lim_{t \rightarrow 0} \left[-4t \log \left(\frac{1}{(4\pi\eta t)^{\frac{3}{2}}} e^{-\frac{[(x-x_0)^2 + (y-y_0)^2 + (z-z_0)^2]}{4\eta t}} \right) \right] \quad (\text{A-3})$$

Next, this limit expression can be decomposed into three parts:

$$\phi^2 = \lim_{t \rightarrow 0} \left[-4t \log \left(\frac{1}{(4\pi\eta t)^{\frac{3}{2}}} \right) \right] + \frac{[(x-x_0)^2 + (y-y_0)^2 + (z-z_0)^2]}{\eta} \quad (\text{A-4})$$

Based on simple tricks on calculus, it can be known that:

$$\lim_{t \rightarrow 0} \left[-4t \log \left(\frac{1}{(4\pi\eta t)^{\frac{3}{2}}} \right) \right] = 0 \quad (\text{A-5})$$

Therefore, the square of geodesic distance in a homogeneous space could be written as the similar equation of a straight line. As we all know, the straight line is the simplest type of geodesics.

$$\phi = \sqrt{\frac{[(x-x_0)^2 + (y-y_0)^2 + (z-z_0)^2]}{\eta}} \quad (\text{A-6})$$

For the case where the metric tensor is non-constant, since the pressure derivative coefficient has been used to construct the metric tensor, the fundamental solution can then be expressed according to the Einstein summation convention as:

$$p_i - p(\vec{r}, t) = \frac{1}{(4\pi t)^{\frac{3}{2}} \sqrt{|g|}} e^{-\frac{g_{ij}x^ix^j}{4\eta t}} \quad (\text{A-7})$$

Similarly, substituting Varadhan's Formula yields the standard geodesic distance formula in differential geometry.

$$\phi = \sqrt{g_{ij}(x^i - x_0^i)(x^j - x_0^j)} \quad (\text{A-8})$$

Appendix B

Algorithm of Uniform Point Distribution Based on Fibonacci Constant

The method to generate uniform-distributed points on a spherical surface could be written as:

$$\begin{aligned}
 z_n &= (2n - 1) / N - 1 \\
 x_n &= \sqrt{1 - z_n^2} \cdot \cos(2\pi nC) \\
 y_n &= \sqrt{1 - z_n^2} \cdot \sin(2\pi nC) \\
 C &= \frac{\sqrt{5}-1}{2} \approx 0.618
 \end{aligned}
 \tag{B-1}$$